Uncertainty-Aware Selection of PSO-Tuned SVMs

A study of model-selection stability on small tabular medical datasets

Research paper — working draft

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This document reports the current implementation and available experiments. It is not a clinical validation study. Dataset provenance, reproducibility, and statistical conclusions must be checked before formal submission.

Abstract

Small datasets can make hyperparameter selection unstable: the candidate with the lowest average cross-validation loss may perform inconsistently across folds. This study examines an uncertainty-aware selection rule for particle-swarm-optimized radial-basis-function support vector machines (RBF-SVMs). Standard selection minimizes mean inner-fold loss; uncertainty-aware selection minimizes mean loss plus a penalty proportional to the sample standard deviation of fold losses. The experiments use nested stratified five-fold cross-validation on Wisconsin Diagnostic Breast Cancer, Pima Indians Diabetes, and Cleveland Heart Disease data, with four preprocessing configurations. A separate threshold-calibration experiment selects a classification threshold from inner out-of-fold scores on outer-training data. A post-hoc sensitivity analysis evaluates penalty values $\lambda \in \{0, 0.25, 0.5, 1, 2\}$ using saved candidate pools. Results are mixed and dataset-dependent. On threshold-calibrated summary comparisons, uncertainty-aware selection improves ROC-AUC in 7 of 12 dataset/preprocessing comparisons, balanced accuracy in 9, and F1 in 9; the absolute inner-to-outer loss gap decreases in 7. These counts are descriptive, not significance tests. The current study supports investigating selection stability, but does not establish universal superiority or clinical usefulness. The small PSO budget, only five outer folds, and unresolved exact provenance of the local Diabetes CSV limit the strength of conclusions.

Keywords: support vector machine; particle swarm optimization; nested cross-validation; model selection; variability penalty; small medical datasets

1. Introduction

Hyperparameter tuning is itself a model-selection problem. If many candidates are compared on a finite validation sample, the selected candidate can benefit from noise in the selection criterion. Cawley and Talbot discuss how selection-criterion variance can lead to overfitting during model selection and bias in performance evaluation [1]. This is especially relevant when the available dataset is small and model performance varies across cross-validation partitions.

This project asks whether a candidate-selection rule that accounts for fold-to-fold variability can improve the generalization behavior of PSO-tuned RBF-SVMs. The proposed uncertainty-aware (UA) rule adds a variability penalty to mean inner-fold loss. Importantly, the current implementation applies this penalty when choosing among candidates in the recorded search history; PSO particle movement and best-position updates still optimize mean inner-fold loss. Therefore, the method is a variability-aware selection rule over a PSO-generated candidate pool, not a PSO optimizer whose dynamics directly optimize the penalty.

Research question: Does selecting PSO-generated RBF-SVM candidates using both mean inner-CV loss and fold-to-fold variability yield better outer-fold performance or a smaller inner-to-outer loss gap than standard minimum-mean-loss selection?

2. Contributions and scope

The implemented contribution is an empirical comparison of two selection rules under the same nested-CV setup, combined with preprocessing comparisons and a post-hoc sensitivity analysis over λ. The work is exploratory: it evaluates three public benchmark datasets and does not claim clinical validation, fairness, or real-world diagnostic readiness.

3. Datasets

Dataset	Rows	Predictors	Target handling
Wisconsin Diagnostic Breast Cancer	569	30	Malignant encoded as positive class
Pima Indians Diabetes	768	8	Binary diabetes outcome
Cleveland Heart Disease	303	13	0 = no disease; 1–4 = disease

For Pima diabetes, the implementation treats zero values in glucose, blood pressure, skin thickness, insulin, and BMI as missing while retaining zero pregnancies as valid. Cleveland Heart Disease uses '?' as missing values, with imputation performed within training folds. The local diabetes file matches the familiar 768-row, eight-predictor Pima classification dataset by shape, but its exact download source/version and checksum are not currently established; that provenance should be recorded before publication.

4. Methodology

4.1 Preprocessing

Four configurations are evaluated: baseline median imputation, mean imputation, PCA with median imputation, and truncated SVD with median imputation. Data transformations are placed within scikit-learn pipelines so fitted preprocessing parameters are learned from the relevant training partition rather than the held-out evaluation fold. The project documentation specifies PCA at 95% retained variance and SVD at five components; verify that these settings remain aligned with the exact script version used to generate the final results.

4.2 RBF-SVM and PSO search

The model is an RBF-kernel SVM. PSO searches log10(C) and log10(γ) in [–3, 3], using six particles and four iterations per outer fold and preprocessing configuration. The primary optimization loss is 1 – ROC-AUC. This is a modest computational budget and should be described as such; it may not explore the search space comprehensively.

4.3 Nested cross-validation

The primary design uses stratified five-fold outer cross-validation and five-fold inner cross-validation. The inner loop supplies candidate evaluation for selection; the outer test fold is held out for the final evaluation of that outer split. Performance summaries should report mean and standard deviation across outer folds. Five folds, however, are not independent repeated experiments and provide limited inferential power.

4.4 Selection criteria

Let L■(θ) denote the inner-fold loss for candidate θ on fold i. Standard selection chooses θ* = argminθ mean■ L■(θ). UA selection chooses θ* = argminθ [mean■ L■(θ) + λ·SD■(L■(θ))], where SD is the sample standard deviation and the primary setting is λ = 1. The penalty discourages candidates whose observed performance is unstable across inner folds. Since PSO's movement uses mean loss, λ changes final selection among recorded candidates rather than the PSO trajectory.

4.5 Threshold calibration

A separate experiment estimates inner out-of-fold decision scores on the outer-training portion and chooses a threshold to maximize inner out-of-fold balanced accuracy. The selected threshold is then applied to the outer test fold. Outer-test labels must not influence threshold selection. This calibration can alter class-label metrics such as balanced accuracy, sensitivity, specificity, and F1; ROC-AUC should remain invariant to threshold changes when the underlying decision scores and fitted model are identical.

4.6 Metrics

The reported metrics are ROC-AUC, balanced accuracy, sensitivity, specificity, F1, and the inner-to-outer overtuning gap. The gap is defined as outer loss minus the selected candidate's mean inner loss, with loss = 1 – ROC-AUC. A positive gap means the outer loss was higher than the selected inner mean loss; a negative gap means the reverse. This is a diagnostic and is not, by itself, proof of overtuning.

5. Results

5.1 Threshold-calibrated comparison

Table 2 reports the consistency-checked, threshold-calibrated summaries. Values are means over five outer folds. The standard and UA results should be compared within the same dataset and preprocessing configuration; aggregate win counts are descriptive only.

Wisconsin Diagnostic Breast Cancer

Preprocessing	Method	AUC	Balanced acc.	F1	Gap
baseline median	Standard	0.9925	0.9695	0.9603	0.0025
baseline median	Uncertainty-aware	0.9925	0.9695	0.9603	0.0025
mean imputation	Standard	0.9925	0.9695	0.9603	0.0025
mean imputation	Uncertainty-aware	0.9925	0.9695	0.9603	0.0025
pca median	Standard	0.9925	0.9574	0.9425	0.0023
pca median	Uncertainty-aware	0.9924	0.9630	0.9512	0.0023
svd median	Standard	0.9920	0.9597	0.9455	0.0019
svd median	Uncertainty-aware	0.9926	0.9620	0.9480	0.0013

Pima Indians Diabetes

Preprocessing	Method	AUC	Balanced acc.	F1	Gap
baseline median	Standard	0.8299	0.7310	0.6561	0.0080
baseline median	Uncertainty-aware	0.8416	0.7410	0.6690	-0.0074
mean imputation	Standard	0.8341	0.7425	0.6685	0.0046
mean imputation	Uncertainty-aware	0.8396	0.7436	0.6727	-0.0050
pca median	Standard	0.8409	0.7457	0.6724	-0.0046
pca median	Uncertainty-aware	0.8429	0.7478	0.6756	-0.0082
svd median	Standard	0.8175	0.7226	0.6473	-0.0023
svd median	Uncertainty-aware	0.8162	0.7058	0.6281	-0.0025

Cleveland Heart Disease

Preprocessing	Method	AUC	Balanced acc.	F1	Gap
baseline median	Standard	0.9039	0.8092	0.7929	0.0058
baseline median	Uncertainty-aware	0.9055	0.8238	0.8040	0.0027
mean imputation	Standard	0.9039	0.8092	0.7929	0.0058
mean imputation	Uncertainty-aware	0.9055	0.8238	0.8040	0.0027
pca median	Standard	0.9000	0.8160	0.8043	0.0116
pca median	Uncertainty-aware	0.9017	0.8432	0.8295	0.0082
svd median	Standard	0.8713	0.7726	0.7524	0.0034
svd median	Uncertainty-aware	0.8659	0.7735	0.7616	0.0062

5.2 Descriptive paired comparison

Across the 12 dataset/preprocessing comparisons, UA has higher mean ROC-AUC in 7/12 comparisons, higher balanced accuracy in 9/12, and higher F1 in 9/12. The absolute overtuning gap is smaller in 7/12 comparisons. Ties are not counted as improvements. These are comparisons of summary means, not p-values or proof of statistical significance.

5.3 Lambda sensitivity analysis

The post-hoc sensitivity analysis evaluates $\lambda \in \{0, 0.25, 0.5, 1, 2\}$ using the saved candidate pools and fresh outer-fold model evaluation. It does not rerun PSO particle movement separately for each λ. Thus, it answers a conditional question: how does the final candidate-selection penalty behave among candidates generated by the original search?

λ	Mean AUC	Mean balanced acc.	Mean F1	Mean gap
0.00	0.9059	0.7738	0.6408	0.0035
0.25	0.9072	0.7730	0.6402	0.0020
0.50	0.9070	0.7747	0.6425	0.0020

λ	Mean AUC	Mean balanced acc.	Mean F1	Mean gap
1.00	0.9074	0.7760	0.6501	0.0004
2.00	0.9073	0.7755	0.6477	-0.0003

The values in Table 3 are unweighted averages across the 12 dataset/preprocessing group means in the saved sensitivity summary. They are descriptive aggregates and can mask heterogeneity between datasets. $\lambda = 1$ should not be described as universally optimal: any final choice of λ should be made within training/validation data or supported by a pre-specified rationale, not chosen after examining outer-test outcomes.

6. Discussion

The results show that the effect of uncertainty-aware selection depends on the dataset and preprocessing configuration. The near-ceiling breast-cancer AUC leaves little room for AUC improvement, although threshold-dependent metrics can still vary. Diabetes shows more variable outcomes and some positive UA comparisons, but SVD is not consistently helped. Heart-disease results also vary by preprocessing. The correct interpretation is therefore conditional: the penalty is a plausible stability-aware selection strategy, but these experiments do not establish a universal advantage.

The overtuning gap needs careful interpretation. It compares a selected inner-CV estimate with outer-fold loss, but these quantities arise from different samples and may be affected by finite-sample variation. A smaller average gap is not by itself proof that the selection method reduces overfitting. Fold-level paired differences and additional repetitions would strengthen the evidence.

7. Limitations and threats to validity

- Small and heterogeneous benchmark collection: three datasets do not establish broad generalization.
- Limited search budget: six particles and four iterations may not adequately explore C and γ.
- Five outer folds: estimates have uncertainty and provide limited statistical power.
- Post-hoc λ analysis: sensitivity is conditional on the saved candidate pools and is not a separate PSO run for each λ.
- Local Diabetes CSV provenance: exact URL/version/checksum remains to be documented.
- Potential preprocessing sensitivity: zero-as-missing handling is a domain assumption and should be explicitly justified and tested.
- Threshold calibration: must be reported separately from default-threshold results and verified to use only outer-training data.
- No external validation or clinical assessment: these are benchmark experiments, not a medical device evaluation.

8. Conclusion

This study implements and examines a variability-penalized candidate-selection rule for PSO-generated RBF-SVMs within nested cross-validation. The empirical findings are mixed and depend on dataset and preprocessing. The present evidence motivates further evaluation but does not establish that UA selection always outperforms standard selection. The next strongest steps are to lock down dataset provenance, report fold-level paired differences, run a pre-specified repeated nested-CV study or additional independent datasets, justify the PSO budget, and ensure that the final manuscript's claims match the implemented method.

9. Reproducibility checklist

Before formal submission: (1) record each raw dataset's source URL, version/date, and SHA-256 checksum; (2) preserve exact Python/package versions and random seeds; (3) verify preprocessing settings against the final scripts; (4) keep default-threshold and calibrated experiments separate; (5) include fold-level means/standard deviations and paired differences; (6) do not claim significance unless a justified analysis supports it; (7) rerun from a clean environment and confirm generated files; (8) state explicitly that λ applies to final selection over the PSO history.

References

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